

Principal Component Analysis

Readings: Zelterman, Sections 8.1–8.4;
Izenman, Sections 7.1–7.2

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Agenda

Background

Finding Principal Components

Principal Component Analysis in Practice

Learning Objectives

- ▶ **Explain the purpose of PCA:** describe how principal components reduce dimensionality while retaining variation in multivariate data
- ▶ **Construct principal components:** use eigenvalues and eigenvectors of a covariance or correlation matrix to define principal components and quantify variance explained
- ▶ **Apply PCA in practice:** determine how many components to retain and interpret component weights, scores, and scree plots
- ▶ **Connect PCA to spatio-temporal applications:** recognize empirical orthogonal function (EOF) analysis as PCA applied to large spatial fields

PCA: The Big Picture

Goal: Can we replace many correlated variables by a few new variables while preserving most of the variation in the data?

1. Correlated data

Many variables may contain overlapping information.

2. Find directions

PCA identifies orthogonal directions that capture the most variation.

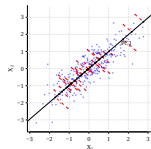
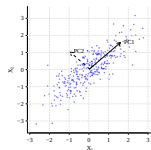
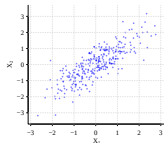
3. Project data onto these directions

Each observation is represented by its PC scores.

4. Retain the leading directions

Keep the first $k < p$ components that capture most of the variation.

Key Idea: PCA finds a new coordinate system in which the first few directions capture as much of the variation in the data as possible.



From a Direction to a Principal Component

PCA finds new directions through the data. For direction k :

1. **Direction:** e_k

A set of weights telling us how to combine the original variables.

2. **New variable:** PC_k

$$PC_k = e_k^\top (X - \mu)$$

The principal component is the resulting linear combination.

3. **Score:** z_{ik}

$$z_{ik} = e_k^\top (x_i - \bar{x})$$

The value of that component for observation i .

4. **Variation:** λ_k

How much the scores vary along direction k .

Keep the roles straight

The **direction** describes the variables; a **score** describes one observation.

History

- Karl Pearson (1901): find low-dimensional lines and planes that best represent multivariate data
- Harold Hotelling (1933): construct a smaller set of variables that dominant variation in the original variables

[III. *On Lines and Planes of Closest Fit to Systems of Points in Space.* By KARL PEARSON, F.R.S., University College, London *.

(1) IN many physical, statistical, and biological investigations it is desirable to represent a system of points in plane, three, or higher dimensioned space by the "best-fitting" straight line or plane. Analytically this consists in taking

$$y = a_0 + a_1x, \quad \text{or} \quad z = a_0 + a_1x + b_1y,$$

$$\text{or} \quad z = a_0 + a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n,$$

where $y, z, x, x_1, x_2, \dots, x_n$ are variables, and determining the "best" values for the constants $a_0, a_1, b_1, a_2, a_3, \dots, a_n$ in relation to the observed corresponding values of the variables. In nearly all the cases dealt with in the text-books of least squares, the variables on the right of our equations are treated as the independent, those on the left as the dependent variables. The result of this treatment is that we get one straight line or plane if we treat some one variable as independent, and a quite different one if we treat another variable as the independent variable. There is no paradox

ANALYSIS OF A COMPLEX OF STATISTICAL VARIABLES INTO PRINCIPAL COMPONENTS¹

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1. INTRODUCTION

Consider n variables attaching to each individual of a population. These statistical variables $x_1, x_2, \dots, x_n$ might for example be scores made by school children in tests of speed and skill in solving arithmetical problems or in reading; or they might be various physical properties of telephone poles, or the rates of exchange among various currencies. The x 's will ordinarily be correlated. It is natural to ask whether some more fundamental set of independent variables exists, perhaps fewer in number than the x 's, which determine the values the x 's will take. If $\gamma_1, \gamma_2, \dots$ are such variables, we shall then have a set of relations of the form

$$x_i = f_i(\gamma_1, \gamma_2, \dots) \quad (i = 1, 2, \dots, n) \quad (1)$$

Quantities such as the γ 's have been called mental factors in recent psychological literature. However in view of the prospect of application of these ideas outside of psychology, and the conflicting usage attaching to the word "factor" in mathematics, it will be better simply to call the γ 's *components* of the complex depicted by the tests.

PCA: From Many Variables to a Few Features

Main idea. PCA replaces p correlated variables with $k < p$ **uncorrelated linear combinations** that preserve as much variation as possible.

$$\underbrace{(X_1, \dots, X_p)}_{p \text{ original variables}} \implies \underbrace{(PC_1, \dots, PC_k)}_{k \text{ principal components}} \quad (k < p)$$

Why use PCA?

- ▶ Dimension reduction / compression
- ▶ Visualize high-dimensional data
- ▶ Uncover dominant patterns
- ▶ Create features for downstream analysis

PCA is feature construction, not feature selection.

Multivariate Data

We display a multivariate data that contains n units on p variables using a matrix

$$\mathbf{X} = \begin{pmatrix} X_{1,1} & X_{2,1} & \cdots & X_{p,1} \\ X_{1,2} & X_{2,2} & \cdots & X_{p,2} \\ \vdots & \cdots & \ddots & \vdots \\ X_{1,n} & X_{2,n} & \cdots & X_{p,n} \end{pmatrix}$$

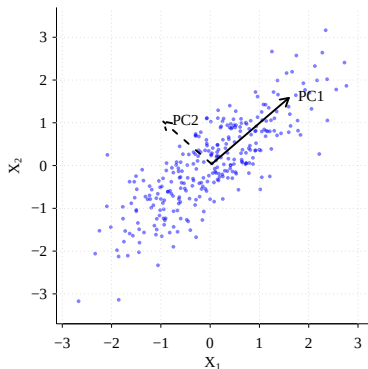
Summary Statistics

► **Mean Vector:** $\bar{\mathbf{X}} = (\bar{X}_1, \bar{X}_2, \dots, \bar{X}_p)^T$, where
$$\bar{X}_j = \frac{\sum_{i=1}^n X_{j,i}}{n}, \quad j = 1, \dots, p$$

► **Covariance Matrix:** $\Sigma = \{\sigma_{ij}\}_{i,j=1}^p$, where
 $\sigma_{ii} = \text{Var}(X_i)$, $i = 1, \dots, p$ and $\sigma_{ij} = \text{Cov}(X_i, X_j)$, $i \neq j$

How do we find the **principal components**?

PCA Finds a New Coordinate System



PC1

Direction along which the projected data have the **greatest variation**.

PC2

An **orthogonal** direction that captures the greatest remaining variation.

More generally: Each new PC direction is orthogonal to the previous ones and captures as much remaining variation as possible.

Geometric View of PCA

PCA rotates the original coordinate system (X_1, X_2) to new orthogonal axes (PC_1, PC_2) aligned with the dominant directions of variation.

Finding Principal Components

Principal Components (PCs) are uncorrelated **linear combinations** $PC_1, PC_2, \dots, PC_p$ determined sequentially, as follows:

1. The first PC is the linear combination $PC_1 = \mathbf{c}_1^\top \mathbf{X} = \sum_{i=1}^p c_{1i} X_i$ that maximizes $\text{Var}(\tilde{X}_1)$ subject to $\mathbf{c}_1^\top \mathbf{c}_1 = 1$
2. The second PC is the linear combination $PC_2 = \mathbf{c}_2^\top \mathbf{X} = \sum_{i=1}^p c_{2i} X_i$ that maximizes $\text{Var}(\tilde{X}_2)$ subject to $\mathbf{c}_2^\top \mathbf{c}_2 = 1$ and $\mathbf{c}_2^\top \mathbf{c}_1 = 0$
- $\vdots$
3. The p_{th} PC is the linear combination $PC_p = \mathbf{c}_p^\top \mathbf{X} = \sum_{i=1}^p c_{pi} X_i$ that maximizes $\text{Var}(\tilde{X}_p)$ subject to $\mathbf{c}_p^\top \mathbf{c}_p = 1$ and $\mathbf{c}_p^\top \mathbf{c}_k = 0, \forall k < p$

Finding PCs from the Covariance Matrix

Let Σ , the covariance matrix of $\mathbf{X}$, have eigenvalue-eigenvector pairs $(\lambda_i, \mathbf{e}_i)_{i=1}^p$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$

- The k_{th} principal component is given by

$$PC_k = \mathbf{e}_k^\top \mathbf{X} = e_{k1}X_1 + e_{k2}X_2 + \dots e_{kp}X_p$$

$\Rightarrow$ we can perform a single matrix operation to get the coefficients to form all the PCs!



$$\text{Var}(PC_k) = \lambda_k, \quad k = 1, \dots, p$$

Moreover $\text{Var}(PC_1) \geq \text{Var}(PC_2) \geq \dots \geq \text{Var}(PC_p) \geq 0$



$$\text{Cov}(PC_j, PC_k) = 0, \quad \forall j \neq k$$

$\Rightarrow$ different PCs are **uncorrelated** with each other

Why Do Eigenvectors Appear in PCA?

For any direction $\mathbf{c}$ with $\mathbf{c}^\top \mathbf{c} = 1$, the variance of the projected data is

$$\text{Var}(\mathbf{c}^\top \mathbf{X}) = \mathbf{c}^\top \mathbf{\Sigma} \mathbf{c}.$$

Thus, the first PC direction solves

$$\max_{\mathbf{c}^\top \mathbf{c} = 1} \mathbf{c}^\top \mathbf{\Sigma} \mathbf{c}.$$

Using a Lagrange multiplier,

$$L(\mathbf{c}, \lambda) = \mathbf{c}^\top \mathbf{\Sigma} \mathbf{c} - \lambda(\mathbf{c}^\top \mathbf{c} - 1).$$

Setting the derivative with respect to $\mathbf{c}$ equal to zero gives

$$2\mathbf{\Sigma} \mathbf{c} - 2\lambda \mathbf{c} = 0 \quad \implies \quad \boxed{\mathbf{\Sigma} \mathbf{c} = \lambda \mathbf{c}}.$$

Key Connection

The direction that maximizes variance is the **eigenvector associated with the largest eigenvalue** of $\mathbf{\Sigma}$.

PCA and Proportion of Variance Explained

- It can be shown that

$$\sum_{i=1}^p \text{Var}(PC_i) = \lambda_1 + \lambda_2 + \cdots + \lambda_p = \sum_{i=1}^p \text{Var}(X_i)$$

- The proportion of the total variance associated with the k_{th} principal component is given by

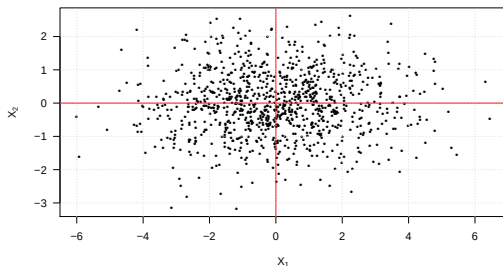
$$\frac{\lambda_k}{\lambda_1 + \lambda_2 + \cdots + \lambda_p}$$

- If the first k PCs explain a large share of total variance (e.g., 80–90%), we can use these $k < p$ uncorrelated components as a lower-dimensional summary of the data

Toy Example 1

Suppose we have $\mathbf{X} = (X_1, X_2)^T$ where $X_1 \sim N(0, 4)$, $X_2 \sim N(0, 1)$ are independent

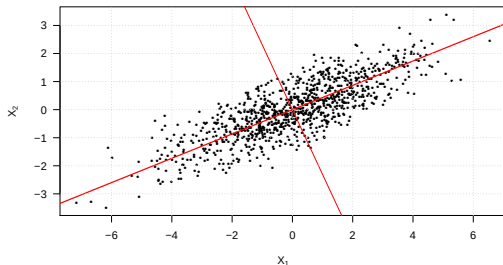
- ▶ Total variation = $\text{Var}(X_1) + \text{Var}(X_2) = 5$
- ▶ X_1 axis explains 80% of total variation
- ▶ X_2 axis explains the remaining 20% of total variation



Toy Example 2

Suppose we have $\mathbf{X} = (X_1, X_2)^T$ where $X_1 \sim N(0, 4)$, $X_2 \sim N(0, 1)$ and $\text{Cor}(X_1, X_2) = 0.8$

- ▶ Total variation
 $= \text{Var}(X_1) + \text{Var}(X_2) = \text{Var}(PC_1) + \text{Var}(PC_2) = 5$
- ▶ $PC_1 = .9175X_1 + .3975X_2$ explains 93.9% of total variation
- ▶ $PC_2 = .3975X_1 - .9176X_2$ explains the remaining 6.1% of total variation



Should We Standardize?

Covariance PCA

PCA of Σ

- ▶ Uses variables on their **original scales**
- ▶ Preserves differences in variance
- ▶ Prefer when scales are comparable and variance magnitudes are **meaningful**

Correlation PCA

PCA of R

- ▶ Equivalent to PCA on **standardized variables**
- ▶ Gives each variable unit variance
- ▶ Prefer when variables have substantially different **units or scales**

Key Modeling Choice

Standardization is not automatic—it changes the question PCA is answering.

PCA in Practice

A practical workflow for PCA

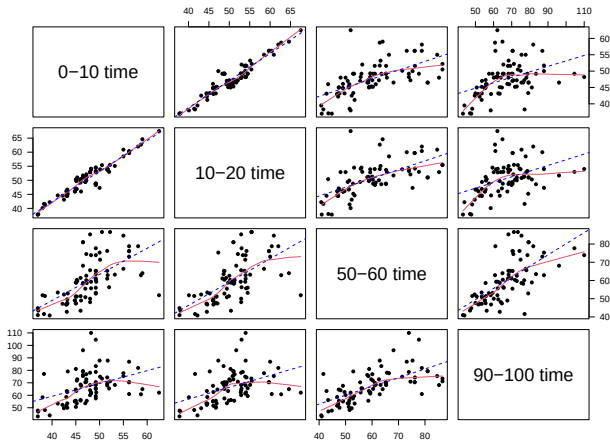
1. **Preprocess:** Center the variables; decide whether to standardize.
2. **Compute PCA:** Compute the principal components.
3. **Choose k :** Check how much variance the first k PCs explain.
4. **Interpret loadings:** Identify which variables define each PC.
5. **Examine scores:** Look for patterns among observations.
6. **Validate:** Check whether the patterns make scientific sense.

Goal

Reduce dimension while retaining an interpretable summary of the data.

Example: Men's 100k Road Race

The data consists of the times (in minutes) to complete successive 10k segments ($p = 10$) of the race. There are 80 racers in total ($n = 80$)



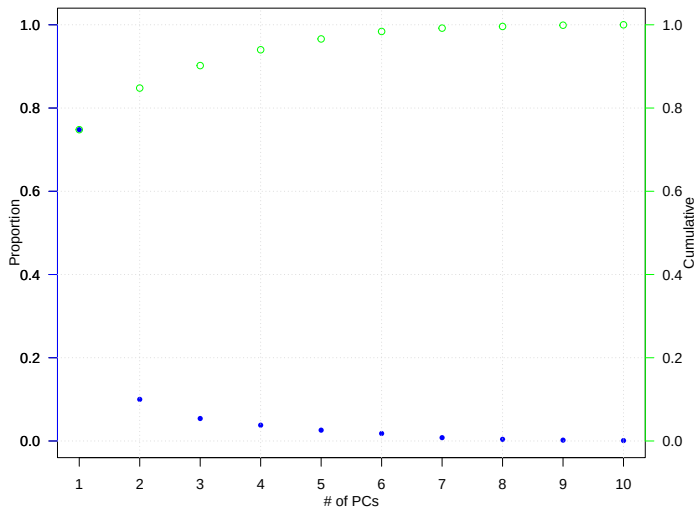
Eigenvalues of $\hat{\Sigma}$

	Eigenvalue	Proportion	Cumulative
PC1	735.77	0.75	0.75
PC2	98.47	0.10	0.85
PC3	53.27	0.05	0.90
PC4	37.30	0.04	0.94
PC5	26.04	0.03	0.97
PC6	17.25	0.02	0.98
PC7	8.03	0.01	0.99
PC8	4.25	0.00	1.00
PC9	2.40	0.00	1.00
PC10	1.29	0.00	1.00

Much of the total variance can be explained by the first three PCs

How Many Components to Retain?

A **scree plot** displays the variance explained by each component

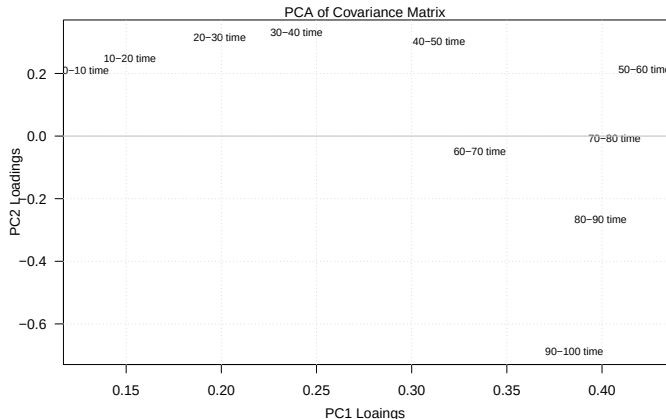


Men's 100k Road Race Loadings

	Comp.1	Comp.2	Comp.3
0-10 time	0.13	0.21	0.36
10-20 time	0.15	0.25	0.42
20-30 time	0.20	0.31	0.34
30-40 time	0.24	0.33	0.20
40-50 time	0.31	0.30	-0.13
50-60 time	0.42	0.21	-0.22
60-70 time	0.34	-0.05	-0.19
70-80 time	0.41	-0.01	-0.54
80-90 time	0.40	-0.27	0.15
90-100 time	0.39	-0.69	0.35

What do these numbers mean?

Visualizing the Weights to Gain Insight



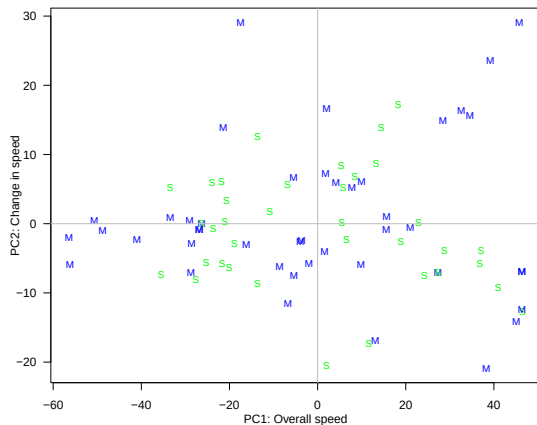
First component: overall race time

Second component: contrast long and short races

Note: The sign of an eigenvector is arbitrary

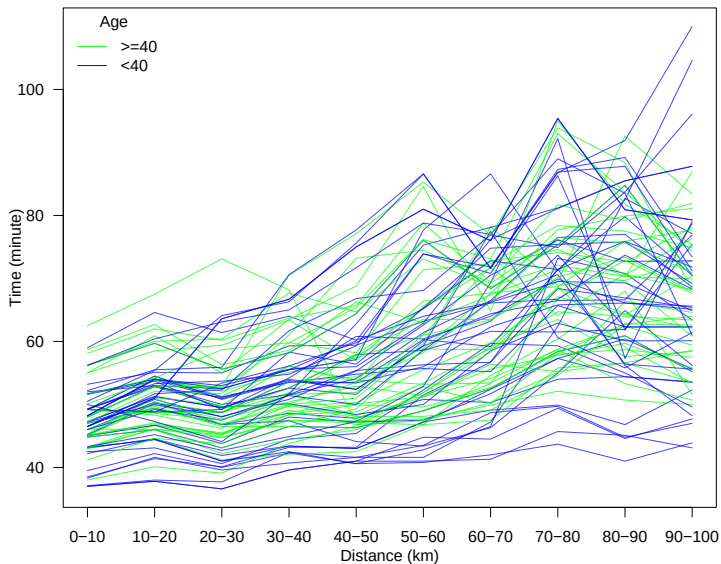
Looking for Patterns

Mature runners: Age < 40 (M); Senior runners: Age ≥ 40 (S)

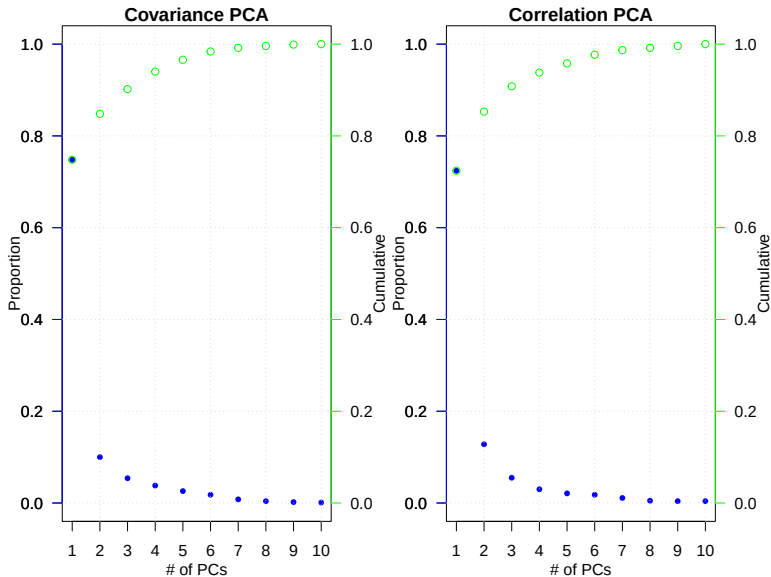


Loadings tell us what a component means. **Scores** here tell us where each observation lies on that component.

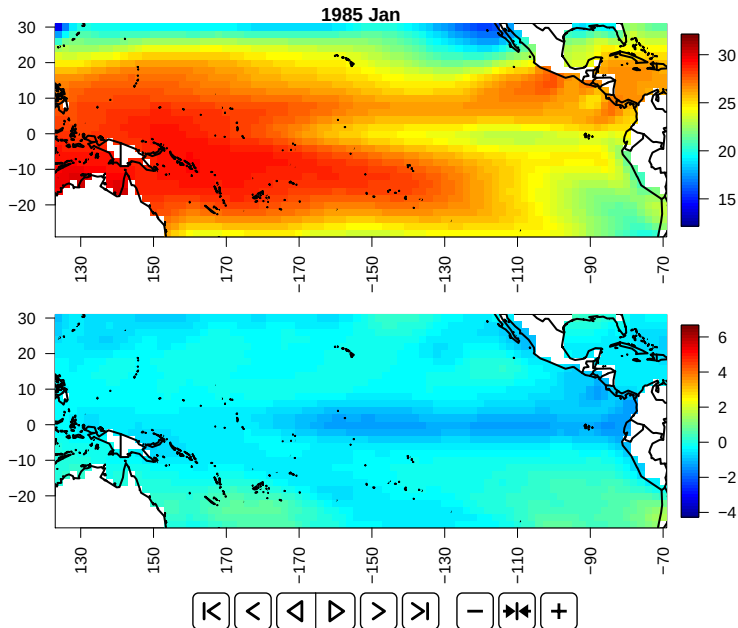
Relating to Original Data: Profile Plot



Correlation PCA versus Covariance PCA



Example: Monthly Sea Surface Temperatures



Sea Surface Temperatures and Anomalies

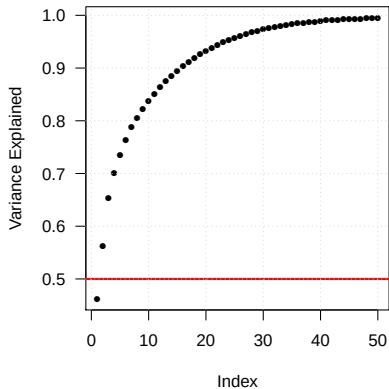
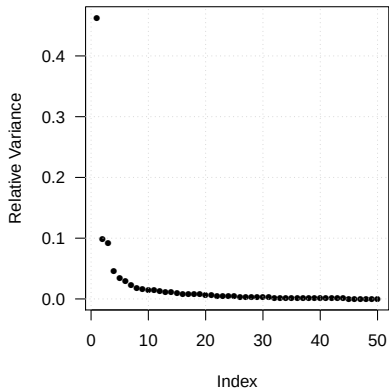
- ▶ The “data” are gridded at a 2° by 2° resolution from $124^\circ E - 70^\circ W$ and $30^\circ S - 30^\circ N$. The dimension of this SST data set is 2303 (number of grid points in space) $\times$ 552 (monthly time series from 1970 Jan. to 2015 Dec.)
- ▶ Sea-surface temperature anomalies are the temperature differences from the climatology (i.e. long-term monthly mean temperatures)
- ▶ We will demonstrate the use of Empirical Orthogonal Function (EOF) analysis to uncover the low-dimensional structure of this spatio-temporal data set

The Empirical Orthogonal Function (EOF) Decomposition

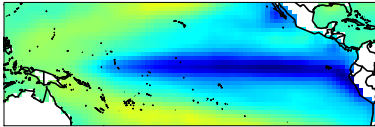
Empirical orthogonal functions (EOFs) are the geophysicist's terminology for the eigenvectors in the eigen-decomposition of an empirical covariance matrix. In its discrete formulation, EOF analysis is simply Principal Component Analysis (PCA). EOFs are usually used

- ▶ To find principal spatial structures
- ▶ To reduce the dimension (spatially or temporally) in large spatio-temporal datasets

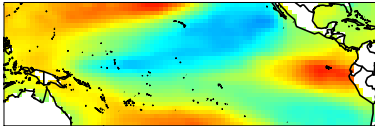
Scree Plot for EOFs



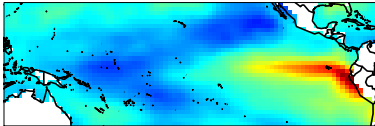
Perform EOF Decomposition and Plot the First Three Modes



EOF1: The classic ENSO pattern

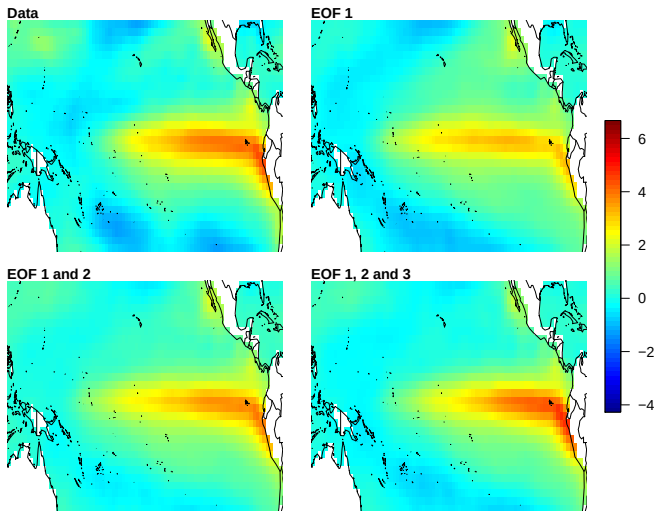


EOF2: A modulation of the center



EOF3: Messing with the coast of SA and the Northern Pacific.

1998 Jan El Niño Event



Low-rank reconstruction: A few leading EOFs can reconstruct the dominant features of the high-dimensional SST field.

PCA: Key Takeaways

1. **PCA finds orthogonal directions that successively maximize variance.**
2. **Eigenvectors define the PC directions; eigenvalues quantify the variance captured by each PC.**
3. **Loadings/weights describe the components; scores locate observations in the new coordinate system.**
4. **The first k PCs provide a low-dimensional approximation** when they capture a sufficiently large proportion of the total variance.
5. **Scaling matters:** covariance-based and correlation-based PCA can produce substantially different results.

Remember

PCA is fundamentally a **representation and dimension-reduction method**, not a predictive model.